INDIAN ECONOMY

LEVEL II

LECTURES

AN OVERVIEW ABOUT THE DAIRY SECTOR IN INDIA - Part II

MODULE - 69

Learning Space

A LOT MORE NEEDS TO BE DONE ON THE YIELD FRONT!

LOOK AT THE AVERAGE MILK YIELD!

- The average milk yield of cattle in India is 1,310 kg per lactation.
- The world average is 2,200 kg.
- In Saudi Arabia and Israel, it is more than 10,000 kg.

WHAT ARE THE MAJOR CONSTRAINTS?

- Scarcity of feed and fodder.
- Inadequate veterinary healthcare services.
- Poor reproductive and productive performance of indigenous animals.
- Wide gap between demand and availability of proven male germ plasm.
- Late sexual maturity of dairy animals.
- Shortage of vaccines and diagnostics.
- Indiscriminate breeding under field conditions.

The per capita milk consumption is now 394 grams per person per day as against the world average of 229 grams per person per day.

THREE IMPORTANT ASPECTS TO ENHANCE YIELD AND KEEP BOVINE ANIMALS IN GOOD HEALTH

1 MAKE AVAILABLE QUALITY FEED AND FODDER

- The availability of quality feed and fodder is a prerequisite for improving reproduction and productivity.
- India is short of dry fodder and green fodder by 12% and 30% respectively.
- The area under fodder cultivation has stagnated at around 9 million hectares, which is around 4.6% of the total cultivable land for the past several decades.

SUB-MISSION ON FODDER AND FEED ANNOUNCED

The Government announced Sub-Mission on Fodder and Feed in view of the shortage of dry fodder, green fodder and concentrate feed.

SUB-MISSION ON FODDER AND FEED

LOOK AT THE STUDY BY INDIAN GRASSLAND AND FODDER RESEARCH INSTITUTE!

- For every 100 kg of feed required, India is short of 23.4 kg of dry fodder, 11.24 kg of green fodder and 28.9 kg of concentrate feed.
- This is one of the primary reasons why Indian livestock's milk productivity is 20%-60% lower than the global average.
- Feed constitutes about 60-70% of the milk production costs.

SUB-MISSION ON FODDER AND FEED

NATIONAL LIVESTOCK MISSION COULD NOT ENSURE SUSTAINABLE FODDER AVAILABILITY!

- The National Livestock Mission was launched in 2014. It focused on supporting farmers in producing fodder from non-forest wasteland, grassland and cultivation of coarse grains.
- This model could not sustain fodder availability due to the lack of backward and forward linkages in the value chain.

NOW, THE SUB-MISSION IS TOWARDS DEVELOPING FODDER ENTREPRENEURS!

- Now the Sub-Mission on Feed and Fodder is intended to focus primarily on assistance towards seed production and the development of feed and fodder entrepreneurs.
- It now provides for 50% direct capital subsidy to the beneficiaries under the Feed and Fodder Entrepreneurship Programme, and 100% subsidy on fodder seed production to identified beneficiaries.

SUB-MISSION ON FODDER AND FEED

HUB AND SPOKE MODEL

- The Sub-Mission intends to create a network of entrepreneurs who will make silage (the hub) and sell them directly to the farmers (the spoke).
- The idea is "funding of the hub" will lead to the development of the spoke.
- The large-scale production of silage will bring down the input costs for farmers since silage is much cheaper than concentrate feed.

SILAGE ... grass or other green fodder compacted and stored in airtight conditions, typically in a silo, without first being dried, and used as animal feed in the winter.

FODDER CROPS WILL BE HIGHLY PROFITABLE!

- Studies have indicated that by growing fodder crops, one can earn Rs. 1.60 by investing Rs. 1 as compared to Rs. 1.20 in the case of common cereals like wheat and rice.
- Private entrepreneurs, Self-Help Groups, Farmer Producer Organizations, dairy cooperative societies, and Section-8 companies (NGOs) can avail themselves of the benefits under the scheme.
- It will provide 50% capital subsidy up to Rs. 50 lakh towards project cost to the beneficiary for infrastructure development and for procuring machinery for value addition in feed such as hay/silage etc.

Not only availability of the fodder, but also, it eliminates middlemen!

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THREE IMPORTANT ASPECTS TO ENHANCE YIELD AND KEEP BOVINE ANIMALS IN GOOD HEALTH

2 TAKING CARE OF ANIMAL HEALTH WITH FOCUS ON VETERINARY HOSPITALS

- Poor animal health and diseases lead to low milk and meat yields. Animals are susceptible to diseases like Foot and Mouth Disease (FMD), Brucellosis or Black Quarter.
- The annual losses due to FMD alone exceeds Rs. 20,000 crore. Goat Plague is causing a loss of about Rs. 180 crore per year.
- The Government launched the National Animal Disease Control Programme (NADCP), aimed at eradicating Foot and Mouth Disease (FMD) and Brucellosis in livestock. It has two components – to control the diseases by 2025 and eradication by 2030.
- The programme has received 100% funding from the Centre, amounting to Rs. 12,652 Cr for five years until 2024.

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A BRIEF ON FOOT AND MOUTH DISEASE

National Animal Disease Control Programme (NADCP) aims at vaccinating over 500 Million Livestock heads including cattle, buffalo, sheep, goats and pigs against the FMD.

FMD is not readily transmissible to humans and is not a public health risk.

- The organism which causes FMD is an aphthovirus. It affects cattle, swine, sheep, goats etc.
- It is not recognised as a zoonotic disease.
- It has high mortality in young animals and death is rare in older animals.
- FMD is found in all excretions and secretions from infected animals.
- It leads to productivity losses.
- Currently, there is no universal FMD vaccine.
- Vaccination can be used to reduce the spread of FMD or protect specific animals. The vaccines are formulated for the specific virus strains present in the country.

A BRIEF ON BRUCELLOSIS

National Animal Disease Control Programme (NADCP) aims to vaccinate 36 Million Female Bovine Calves annually against Brucellosis.

- Brucellosis is caused by a group of bacteria from the genus Brucella. It is a zoonotic disease.
- These bacteria can infect both humans and animals. It rarely spreads from one human to another.
- There is vaccine for animals, but not for humans.
- A variety of animals can contract brucellosis, which includes goats, cattle and dogs. It causes early abortions in animals.
- Reference vaccine is available for one of the Brucella Abortus strain. Any other vaccine should be compared with this reference vaccine.
- People, who eat or drink raw animal products, like unpasteurized milk and unprocessed cheese, as well as raw meat and those who come in contact with animal urine, blood, or tissue may contract the disease.

THREE IMPORTANT ASPECTS TO ENHANCE YIELD AND KEEP BOVINE ANIMALS IN GOOD HEALTH

There is a need for at least one 'flow cytometer' for at least 100 semen centers for rapid and accurate analysis of sperm function.

3 MAINTAINING GENETIC PURITY

- Most important aspect is maintaining genetic purity among indigenous animals.
- This requires semen conservation facilities of indigenous breeds.
- India is the only country with
  - ✓ 50 breeds of cattle
  - ✓ 19 breeds of buffalo
  - ✓ 34 breeds of goat and
  - ✓ 44 breeds of sheep.

TWO IMPORTANT MEASURES TAKEN BY THE GOVERNMENT

ARTIFICIAL INSEMINATION

- The target is to provide doorstep Artificial Insemination services for 300 villages per district in 605 districts from September 2019 to May 2020 under Phase-I.
- The superior female calves would yield around 13.5 lakh tonnes of additional milk per annum after three years.

SEX SORTED SEMEN

- It is one of the breed improvement efforts in cattle and buffaloes, so that farmers get productive animals.
- This can provide only female calves with more than 90% accuracy by using sexed semen.
- This ensures not only increased milk production but also fast-tracking profitability of dairy farmers.

The sexed semen technology is going to be a game changer in breed improvement efforts of India.

LUMPY SKIN DISEASE CAUSED SUBSTANTIAL ECONOMIC LOSSES DURING 2022!

WHAT IS LUMPY SKIN DISEASE?

- Lumpy Skin Disease is caused by the Lumpy Skin Disease Virus (LSDV), which belongs to the genus capripoxvirus. It is part of the poxviridae family - smallpox and monkeypox viruses are also a part of the same family.
- The LSDV has antigenic similarities with the sheeppox virus (SPPV) and the goatpox virus (GTPV).
- It is not a zoonotic virus, meaning the disease cannot spread to humans.

HOW DOES IT SPREAD FROM ANIMAL TO ANIMAL?

- It is spread by vectors like mosquitoes, some biting flies, and ticks and usually affects host animals like cows and water buffaloes.
- At the same time, infected animals shed the virus through oral and nasal secretions which may contaminate common feeding and water troughs.
- Hence, the disease can either spread through direct contact with the vectors or through contaminated fodder and water. It can also spread through animal semen during artificial insemination.

LUMPY SKIN DISEASE

HOW DOES IT AFFECT THE ANIMALS?

HUGE ECONOMIC LOSSES IN 2022

- The disease leads to reduced milk production as the animal becomes weak and also loses appetite due to mouth ulceration.
- The income losses can also be due to poor growth, reduced draught power capacity and reproductive problems associated with abortions, infertility and lack of semen for artificial insemination.

- LSD affects the lymph nodes of the infected animal. It causes the nodes to enlarge and appear like lumps on the skin with diameters varying from 2 to 5 cm.
- The nodules may later turn into ulcers and eventually develop scabs over the skin.
- The other symptoms include
  - ✓ high fever
  - ✓ sharp drop in milk yield
  - ✓ discharge from the eyes and nose
  - ✓ salivation
  - ✓ loss of appetite etc.

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LUMPY SKIN DISEASE

WHAT IS THE INCUBATION PERIOD?

- The incubation period or the time between infection and symptoms is about 28 days as per FAO.
- As per other estimates, it ranges from 4 to 14 days.

WHAT ABOUT THE MORBIDITY?

- The morbidity of the disease varies between 2 to 45% and the mortality or rate of death is less than 10%,
- However, the reported mortality of the outbreak of 2022 in India is up to 15%.

WHERE DID IT ORIGINATE?

- The disease was first observed in Zambia in 1929 and spread to South Asia through Bangladesh in July, 2019 and then reached India in August 2019 with initial cases being detected in Odisha and West Bengal.
- It spread to several regions in India since June, 2022.

LUMPY SKIN DISEASE

IS IT SAFE TO CONSUME THE MILK OF THE CATTLE?

As per Indian Veterinary Research Institute, there is no danger, when people take milk after boiling or without boiling!

- As per FAO, it has not been possible to ascertain the presence of viable and infectious LSDV virus in milk derived from the infected animal.
- However, as large portion of the milk is processed after collection and is either pasteurised or boiled or dried in order to make milk powder, the processes ensure that the virus is inactivated or destroyed.
- Hence, it is safe to consume milk from cattle infected by LSD, as it is a non-zoonotic disease.

LUMPY SKIN DISEASE

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WHAT ABOUT THE VACCINES?

- As per Union Ministry, "Goat Pox Vaccine" is very effective against LSD and is being used across affected States to contain the spread.
- Till the end of 2022, around one crore doses of vaccination have been administered.
- At the same time, two institutes of the ICAR have developed an indigenous vaccine for LSD, which the Centre plans to commercialise and roll out in early 2023.

WHAT ARE THE MEASURES SUGGESTED BY FAO?

- Vaccination of susceptible populations with more than 80% coverage.
- Movement control of bovine animals and quarantining.
- Implementing biosecurity through vector control by sanitising sheds and spraying insecticides.
- Strengthening active and passive surveillance.
- Spreading awareness on risk mitigation among all stakeholders involved.
- Creating large protection and surveillance zones and vaccination zones.

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